Objective Type Questions

13. The difference between two successive odd numbers is
    a) 1 b) 2 c) 3 d) 0

14. The only even prime number is
    a) 4 b) 6 c) 2 d) 0

15. Which of the following numbers is not a prime?
    a) 53 b) 92 c) 97 d) 71

16. The sum of the factors of 27 is
    a) 28 b) 37 c) 40 d) 31

17. The factors of a number are 1, 2, 4, 5, 8, 10, 16, 20, 40 and 80.What is the number?
    a) 80 b) 100 c) 128 d) 160

18. The prime factorisation of 60 is $2 \times 2 \times 3 \times 5$. Any other number which has the same prime factorisation as 60 is
    a) 30 b) 120 c) 90 d) impossible

19. If the number $6354 * 97$ is divisible by 9, then the value $*$ is
    a) 2 b) 4 c) 6 d) 7

20. The number 87846 is divisible by
    a) 2 only b) 3 only c) 11 only d) all of these

1.5 Common Factors

Consider the numbers 45 and 60. Use of divisibility tests will also help us to find the factors of 45 and 60. The factors of 45 are 1,3,5,9,15 and 45 and the factors of 60 are 1,2,3,4,5,6,10,12,15,20, 30 and 60. Here, the common factors of 45 and 60 are 1,3,5 and 15.

As factors of a number are finite, we can think of the **Highest Common Factor** of numbers, shortly denoted as **HCF**.

1.5.1 Highest Common Factor (HCF)

Think about the situation:

Situation 1:

Pavithra plans to celebrate Deepavali by distributing sweets and savouries to the families which cannot afford to buy them. Pavithra's mother gives her 63 athirasams and 42 murukkus. Each family should be given the same number of athirasams and the same number of murukkus. What is the greatest number of families that she can distribute?

Now, Pavithra can tackle this situation by using HCF as given in the following illustration.

6th_01_NUMBERS_T2_EM.indd 12 27-06-2020 04:41:31 PM